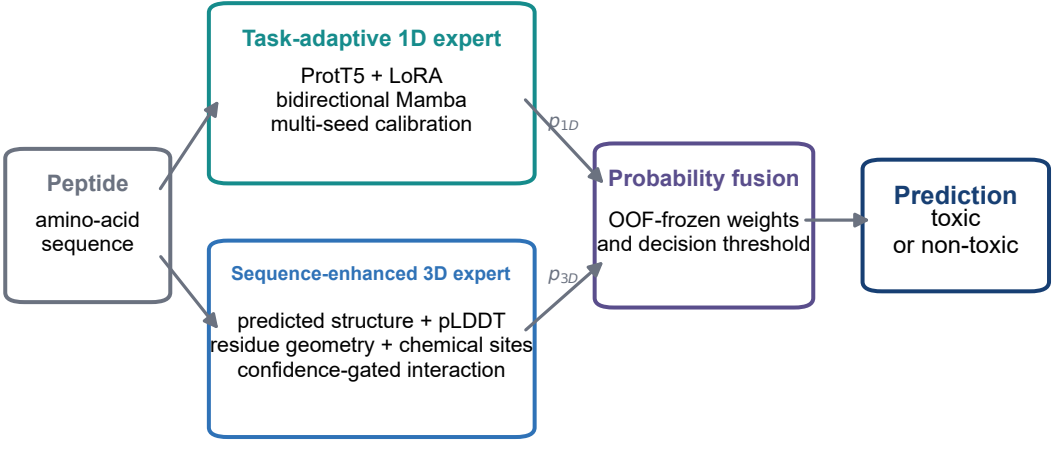
Peptide
amino-acid sequence
Task-adaptive 1D expert
ProtT5 + LoRA
bidirectional Mamba
multi-seed calibration
Sequence-enhanced 3D expert
predicted structure + pLDDT
residue geometry + chemical sites
confidence-gated interaction
p_{1D}
p_{3D}
Probability fusion
OOF-frozen weights and decision threshold
Prediction
toxic
or non-toxic